1.1 Exploratory Data Analysis (EDA)

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Data from 1960 to 2024 on Total Live Births (TLB) and Total Fertility Rates (TFR) was pulled from Singapore Department of Statistics (data.gov.sg). The data was then transformed into a tsibble object with three columns: Year, Total Fertility Rates (TFR) and Total Live Births.

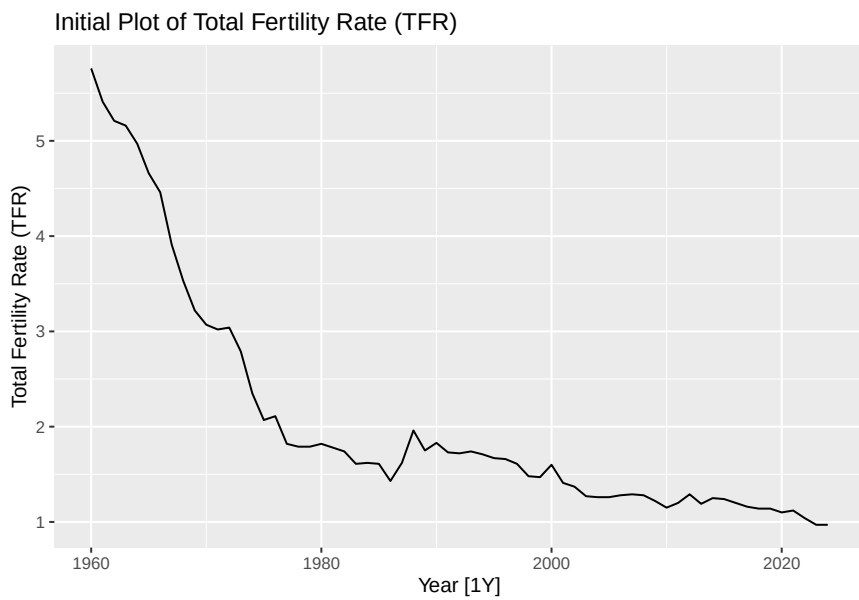
```r
library(fpp3) # time series analysis package
library(tsibble) # tsibble objects
library(lubridate) # for data to work with dates and times
library(feasts)
library(latex2exp)
library(patchwork) # putting 2 plots next to each other

data <- readr::read_csv("BirthsAndFertilityRatesAnnual.csv") # read in the data
BirthAndFertility <- data |>
  pivot_longer( # change rows to column
    cols = `1960`:`2025`,
    names_to = "Year",
    values_to = "Value", # naming it to remember that it needs to be values
    values_transform = list(Value = as.numeric)
  ) |>
  filter(DataSeries %in% c("Total Fertility Rate (TFR)", "Total Live-Births")) |>
  mutate(Year = as.integer(Year)) |>
  filter(Year != 2025) |> # assignment specified until 2024
  pivot_wider( # taking TFR & Total Live-Births into 2 separate columns according to year
  names_from = DataSeries,
  values_from = Value
  )
BirthAndFertility <- as_tsibble(BirthAndFertility, index = Year) # created a tsibble
is_tsibble(BirthAndFertility) # checking it is a tsibble
```

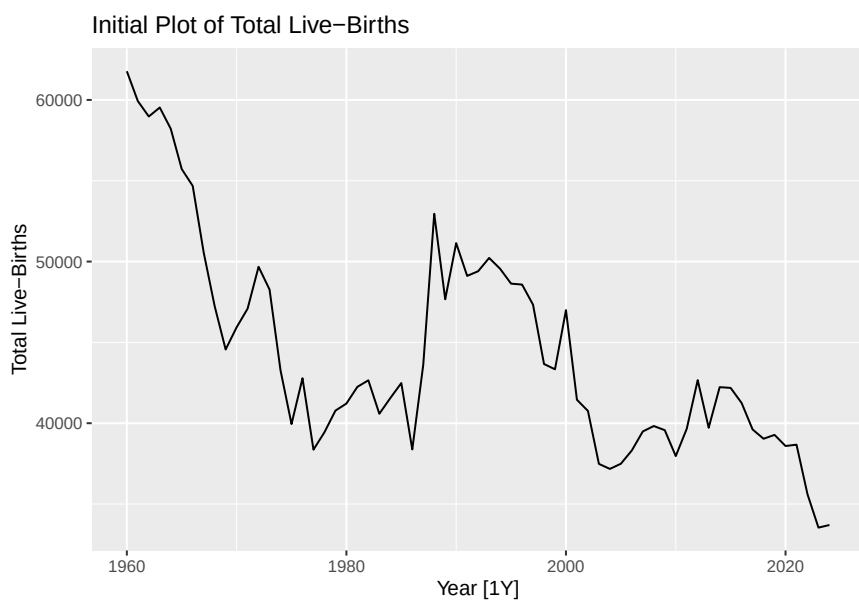
```
## [1] TRUE
```

In this exploratory data analysis, the data was visualised to identify patterns and trends over time. Since the data is yearly, classical decomposition and the X-11 method are not applicable. Once the patterns are identified, appropriate models can be selected, fitted, and evaluated for forecasting.

```r
autoplot(BirthAndFertility, `Total Fertility Rate (TFR)`) +
  labs(title = "Initial Plot of Total Fertility Rate (TFR)")
```



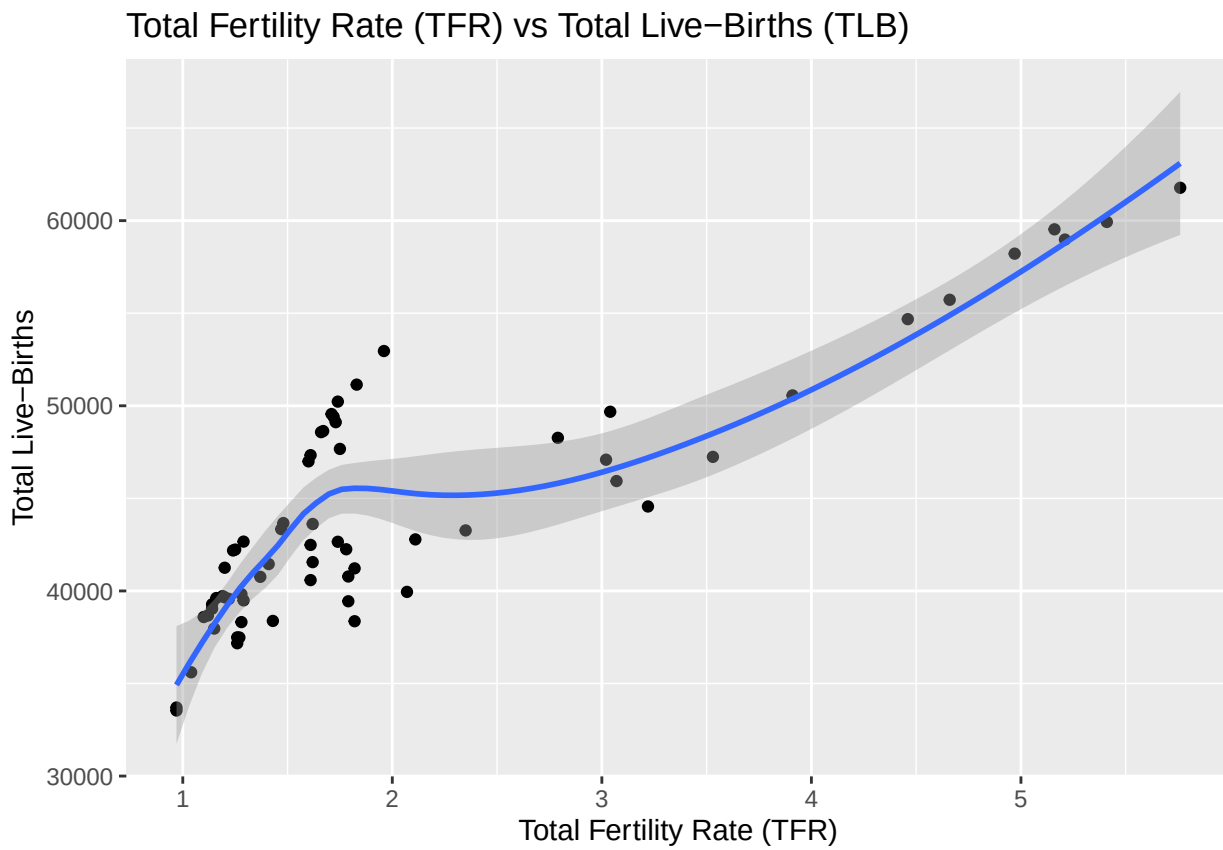

```r
autoplot(BirthAndFertility,`Total Live-Births`)+
  labs(title = "Initial Plot of Total Live-Births")
```



From the initial plots of TFR and TLB, both series show a decreasing trend over time, with TFR exhibiting a more pronounced decline. As the data is yearly, seasonality cannot be determined. While both series may suggest cyclic behaviour, the trend would need to be removed to confirm this.

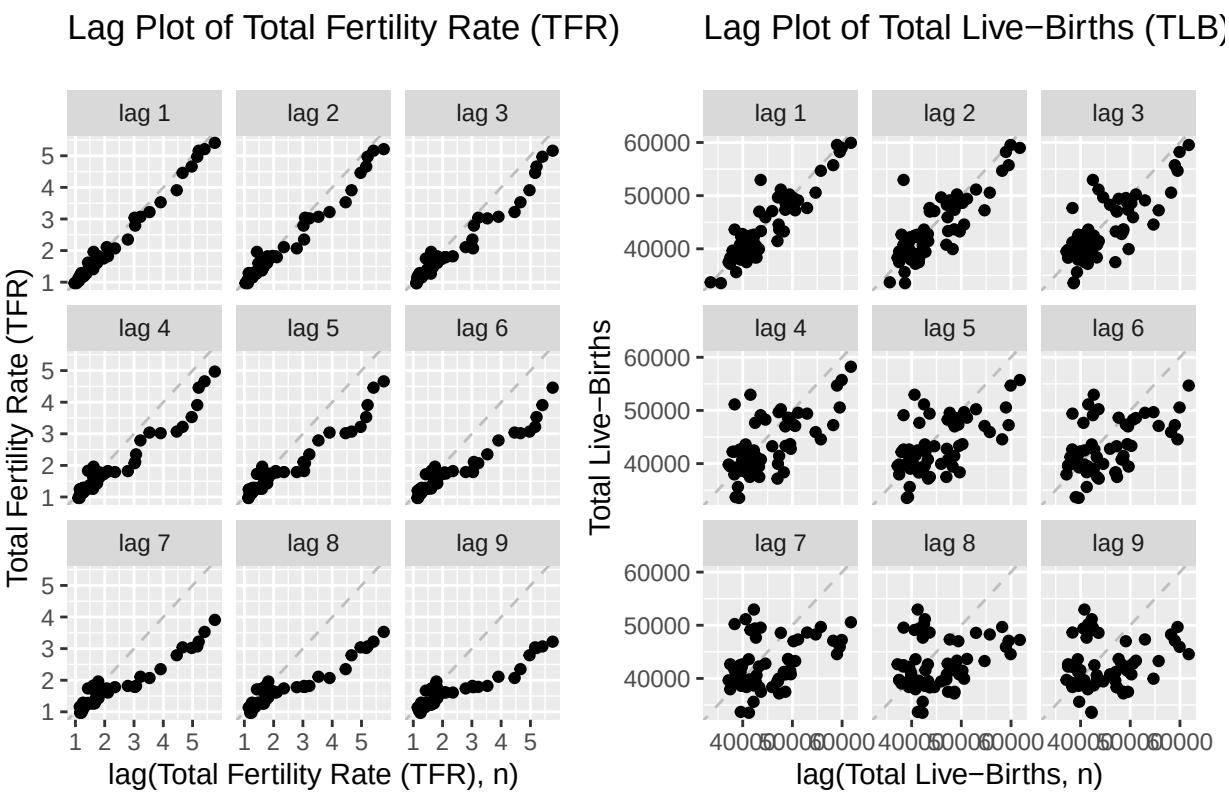
```r
BirthAndFertility |>
  ggplot(aes(x = `Total Fertility Rate (TFR)`, y = `Total Live-Births`)) +
  geom_point() +
  geom_smooth() +
  labs(title = "Total Fertility Rate (TFR) vs Total Live-Births (TLB)")
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



The relationship between TFR and TLB was examined by plotting the two variables against each other. There appears to be a nonlinear positive relationship, suggesting that higher fertility rates are associated with higher numbers of live births, although further analysis is required.

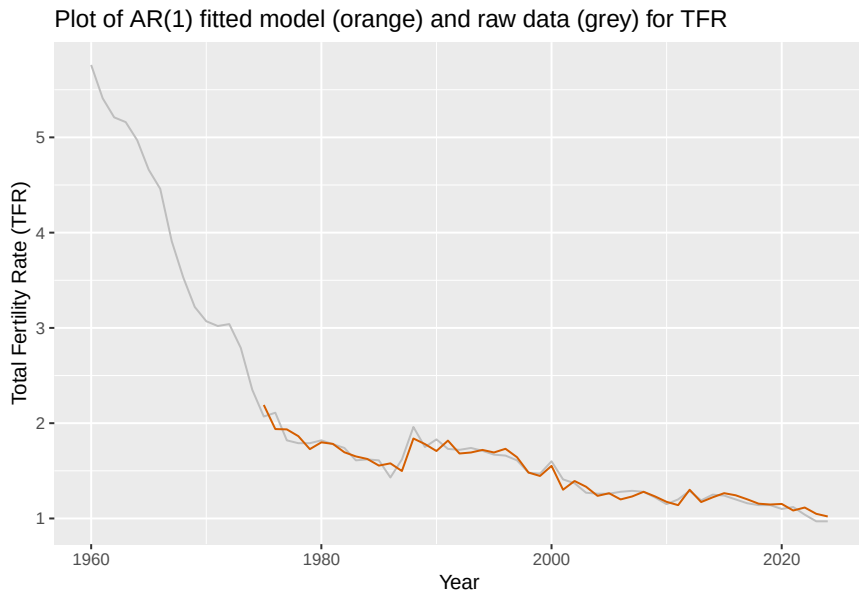
```r
p1 <- BirthAndFertility |>
  gg_lag(`Total Fertility Rate (TFR)`, geom = 'point') +
  labs(title = "Lag Plot of Total Fertility Rate (TFR)")
p2 <- BirthAndFertility |>
  gg_lag(`Total Live-Births`, geom = 'point') +
  labs(title = "Lag Plot of Total Live-Births (TLB)")
p1 + p2 # placing the plots side by side
```



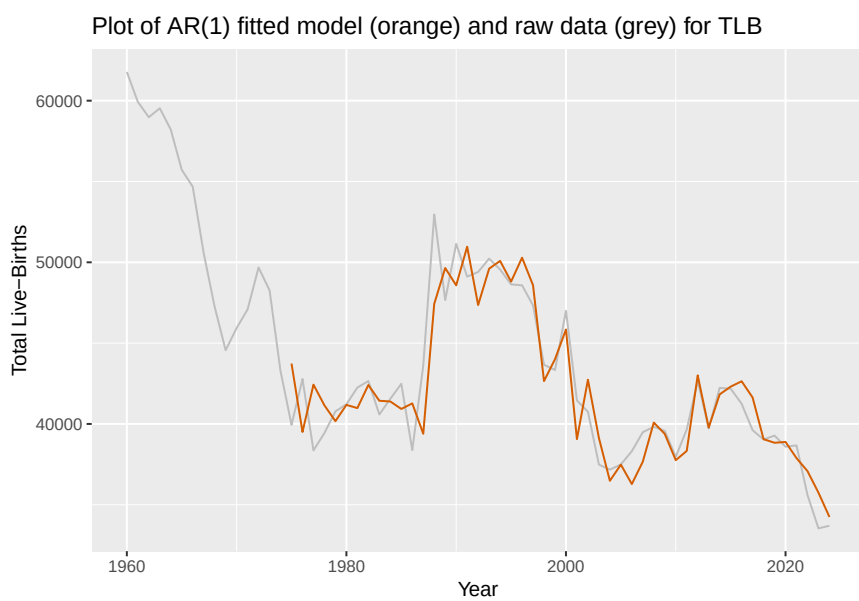
The above lag plots for TFR and TLB indicate stronger positive relationships at lags 1 and 2 suggesting that AR(1) and AR(2) models could be appropriate.

The following fitted AR(1) and AR(2) models (orange lines) follow the TFR and TLB (grey lines) but tends to overshoots and underestimate peaks. Residuals were examined and appear to have no pattern; however the ACF plots show multiple spikes outside the significance bounds indicating the residuals are not white noise. This indicates that an ARIMA model, which can account for trend, may be more appropriate.

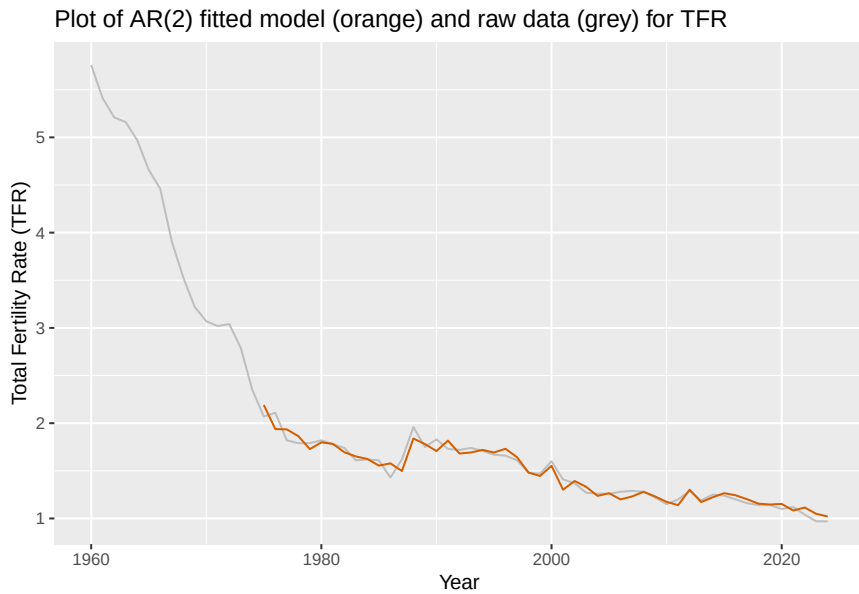
```r
TFR_ar <- BirthAndFertility |> model(AR(`Total Fertility Rate (TFR)`, order = 1)) # fits AR model
TFR_fits <- augment(TFR_ar) # adds columns of data
ggplot(TFR_fits, aes(x = Year)) +
  geom_line(aes(y =`Total Fertility Rate (TFR)`), colour = "grey") + # raw data
  geom_line(aes(y = .fitted), colour = "#D55E00") + #
  labs(title = "Plot of AR(1) fitted model (orange) and raw data (grey) for TFR")
```



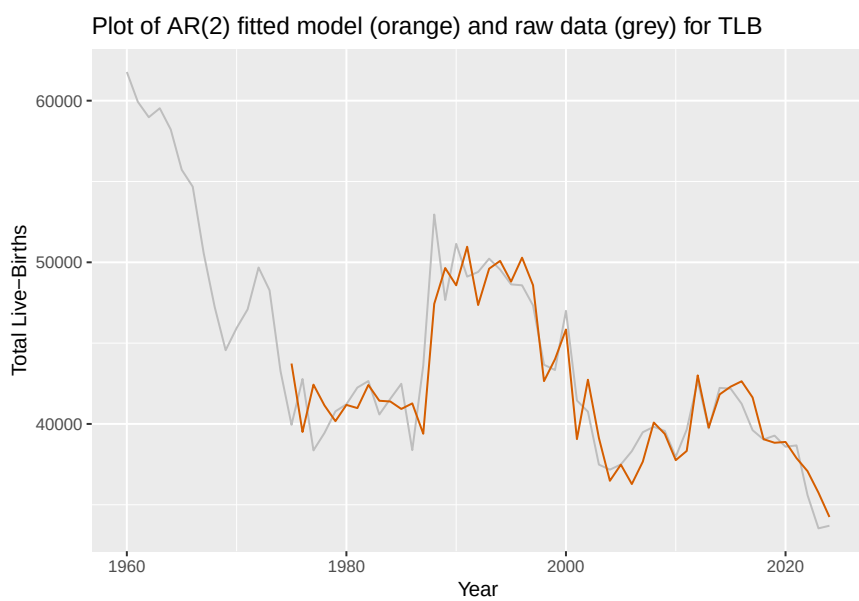

```r
TLB_ar <- BirthAndFertility |> model(AR(`Total Live-Births`, order = 1))
TLB_fits <- augment(TLB_ar)
ggplot(TLB_fits, aes(x = Year)) +
  geom_line(aes(y =`Total Live-Births`), colour = "grey") + # raw data
  geom_line(aes(y = .fitted), colour = "#D55E00") + #
  labs(title = "Plot of AR(1) fitted model (orange) and raw data (grey) for TLB")
```



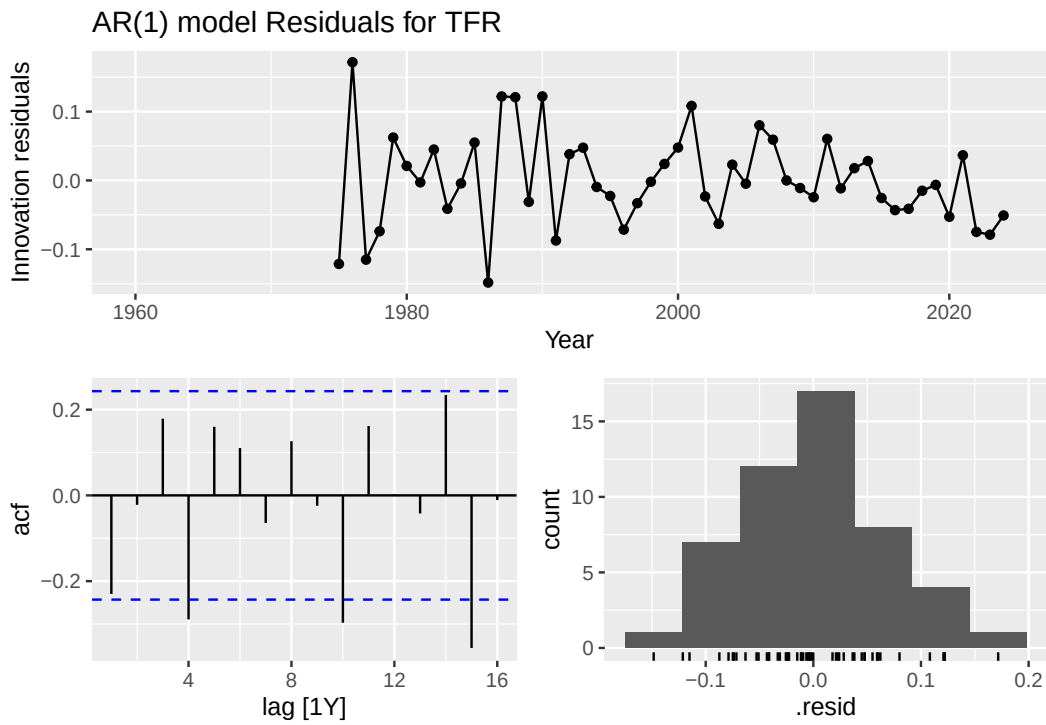
```r
TFR_ar2 <- BirthAndFertility |> model(AR(`Total Fertility Rate (TFR)`, order = 2)) # fits AR model
TFR_fits2 <- augment(TFR_ar2) # adds columns of data
ggplot(TFR_fits2, aes(x = Year)) +
  geom_line(aes(y =`Total Fertility Rate (TFR)`), colour = "grey") + # raw data
  geom_line(aes(y = .fitted), colour = "#D55E00") + #
  labs(title = "Plot of AR(2) fitted model (orange) and raw data (grey) for TFR")
```



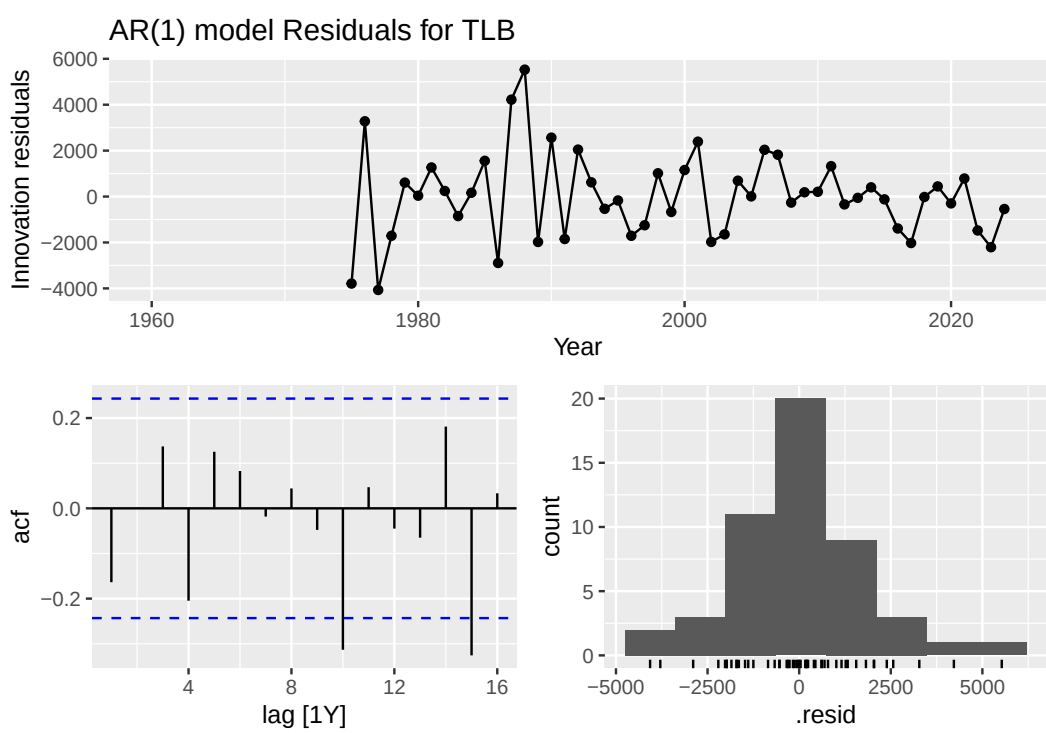

```r
TLB_ar2 <- BirthAndFertility |> model(AR(`Total Live-Births`, order = 2))
TLB_fits2 <- augment(TLB_ar2)
ggplot(TLB_fits2, aes(x = Year)) +
  geom_line(aes(y =`Total Live-Births`), colour = "grey") + # raw data
  geom_line(aes(y = .fitted), colour = "#D55E00") + #
  labs(title = "Plot of AR(2) fitted model (orange) and raw data (grey) for TLB")
```



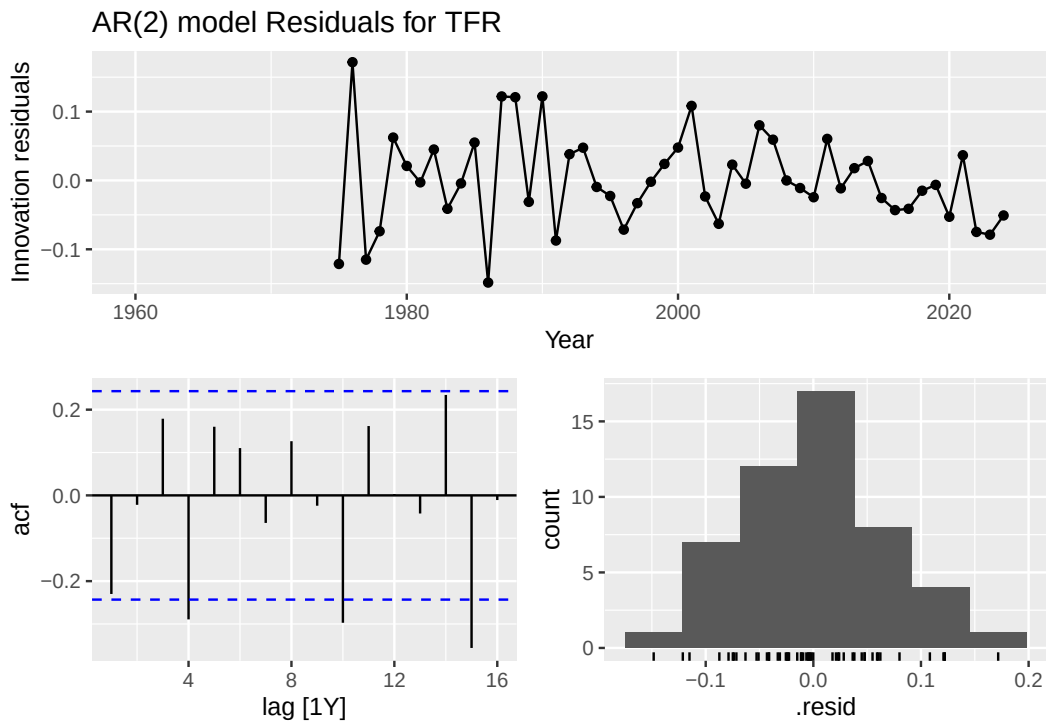
```r
gg_tsresiduals(TFR_ar) + # check residuals
  labs(title = "AR(1) model Residuals for TFR")
```



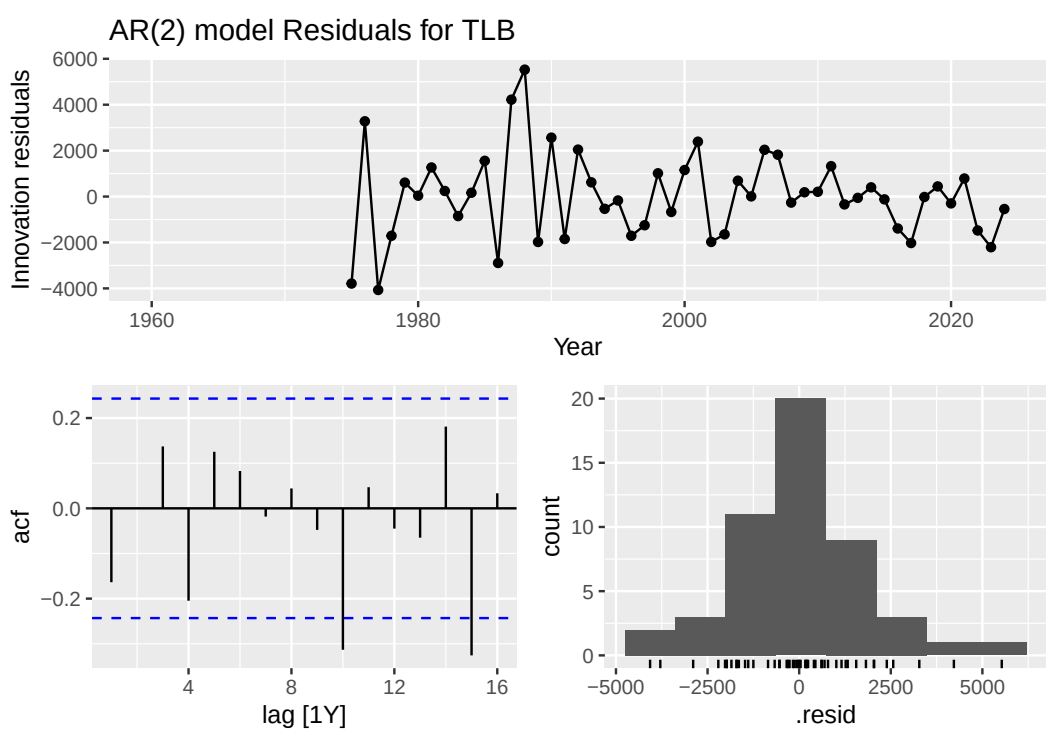

```r
gg_tsresiduals(TLB_ar) +
  labs(title = "AR(1) model Residuals for TLB")
```



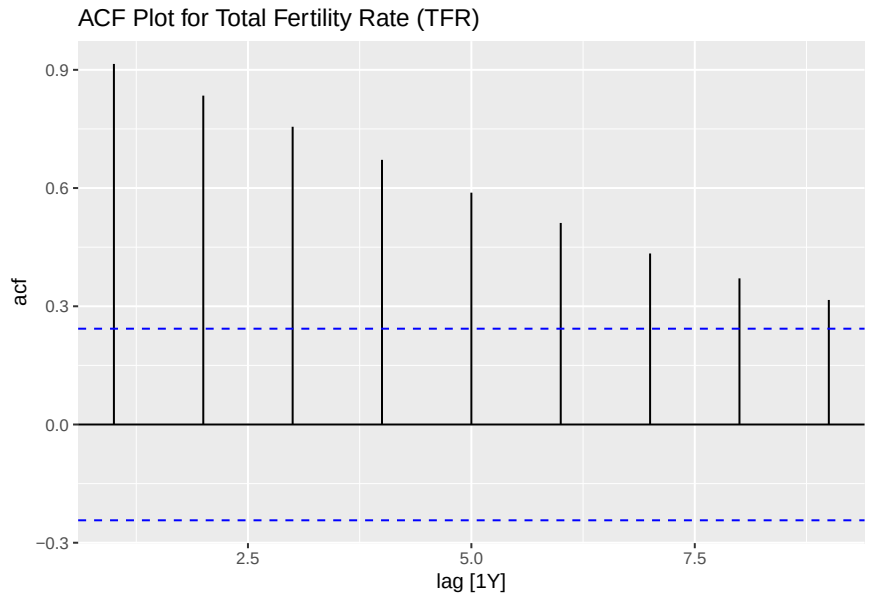
```r
gg_tsresiduals(TFR_ar2) + # check residuals
  labs(title = "AR(2) model Residuals for TFR")
```



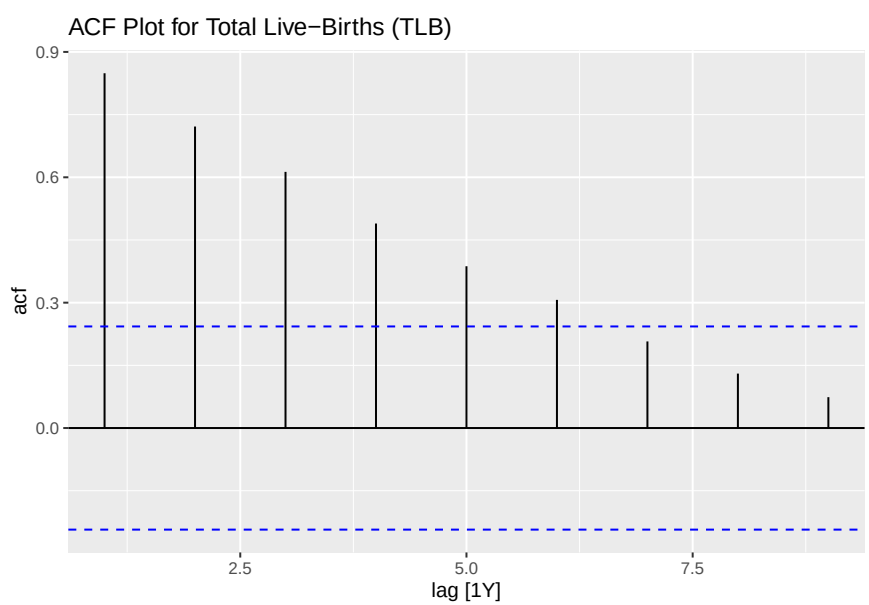

```r
gg_tsresiduals(TLB_ar2) +
  labs(title = "AR(2) model Residuals for TLB")
```



```r
BirthAndFertility |>
  ACF(`Total Fertility Rate (TFR)`, lag_max = 9) |>
  autoplot() +
  labs(title = "ACF Plot for Total Fertility Rate (TFR)")
```

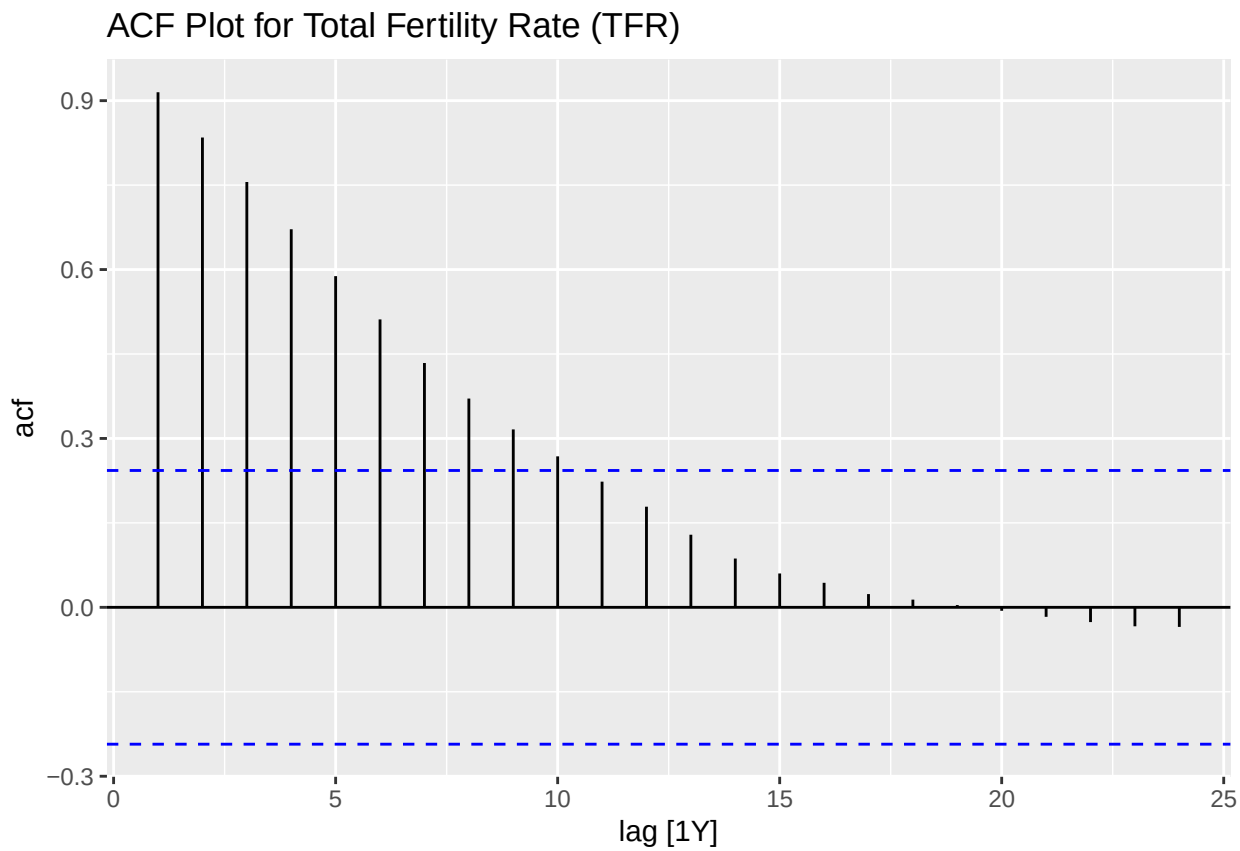



```r
BirthAndFertility |>
  ACF(`Total Live-Births`, lag_max = 9) |>
  ggplot2::autoplot() +
  labs(title = "ACF Plot for Total Live-Births (TLB)")
```



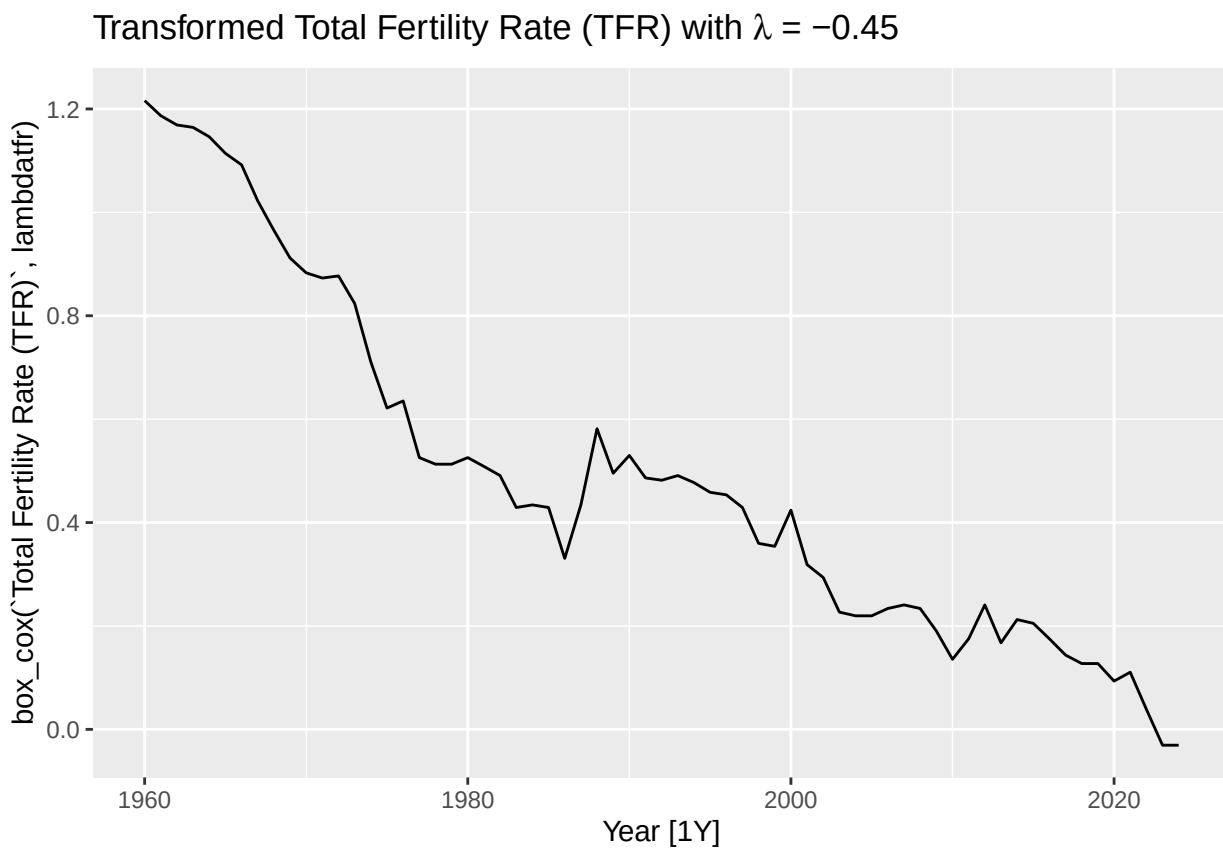
The ACF for TFR shows significant positive autocorrelations from lags 1 to 9, while TLB shows significant positive autocorrelations from lags 1 to 6, with a slow decrease in the ACF as the lag increases. This suggests that neither series is white noise, as there are multiple spikes outside the significance bounds. This slow decrease indicates the presence of a trend and consistent with a non-stationary series. The absence of a sharp cut-off suggests that an autoregressive (AR) model may be suitable for this series.

```r
BirthAndFertility |>
  ACF(`Total Fertility Rate (TFR)`, lag_max = 24) |>
  autoplot() +
  labs(title = "ACF Plot for Total Fertility Rate (TFR)")
```



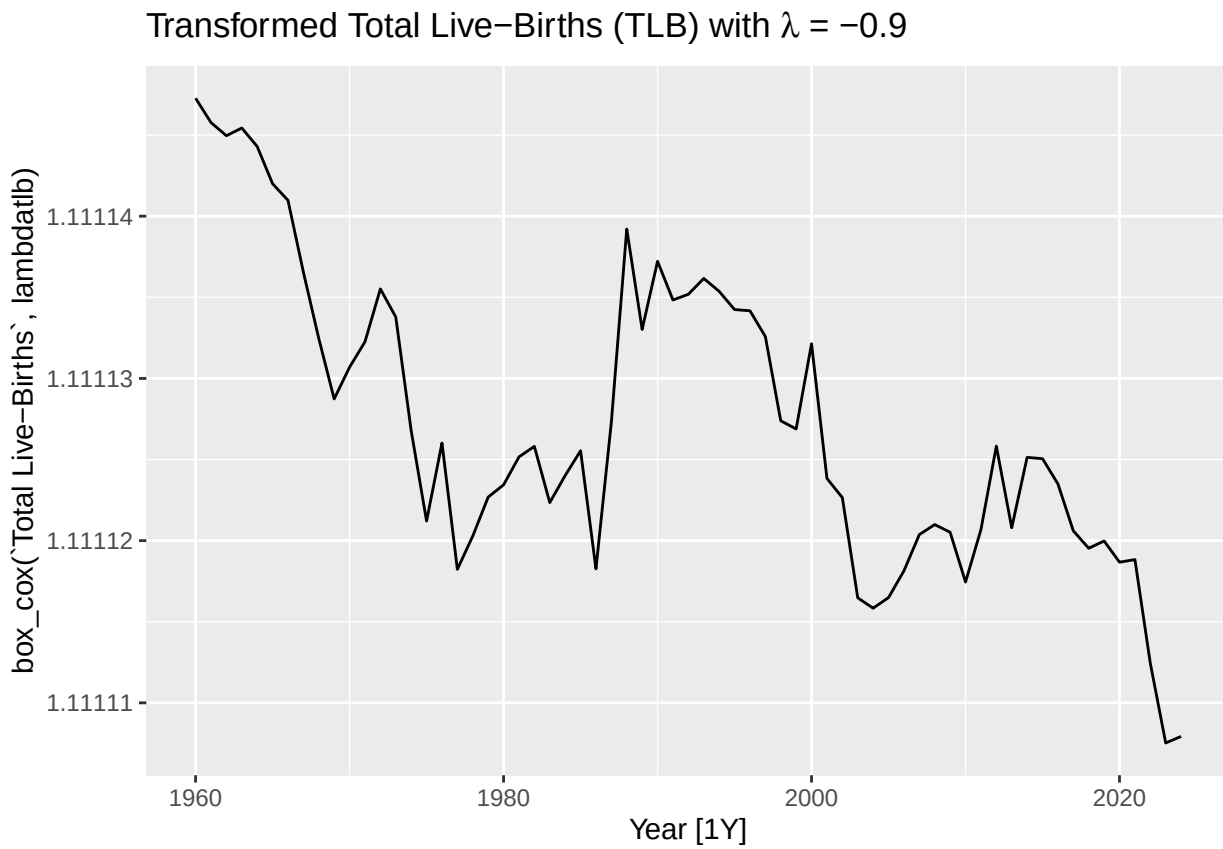
An extended ACF plot for TFR confirms that significant positive autocorrelations persist up to lag 10.

```r
lambdatfr <- BirthAndFertility |>
  features(`Total Fertility Rate (TFR)`, features = guerrero) |> # Guerrero method
  pull(lambda_guerrero) # Box-Cox transformation parameter = -0.44514
BirthAndFertility |>
  autoplot(box_cox(`Total Fertility Rate (TFR)`, lambdatfr)) +
  labs(title = latex2exp::TeX(paste0(
    "Transformed Total Fertility Rate (TFR) with $\\lambda$ = ", round(lambdatfr,2))))
```



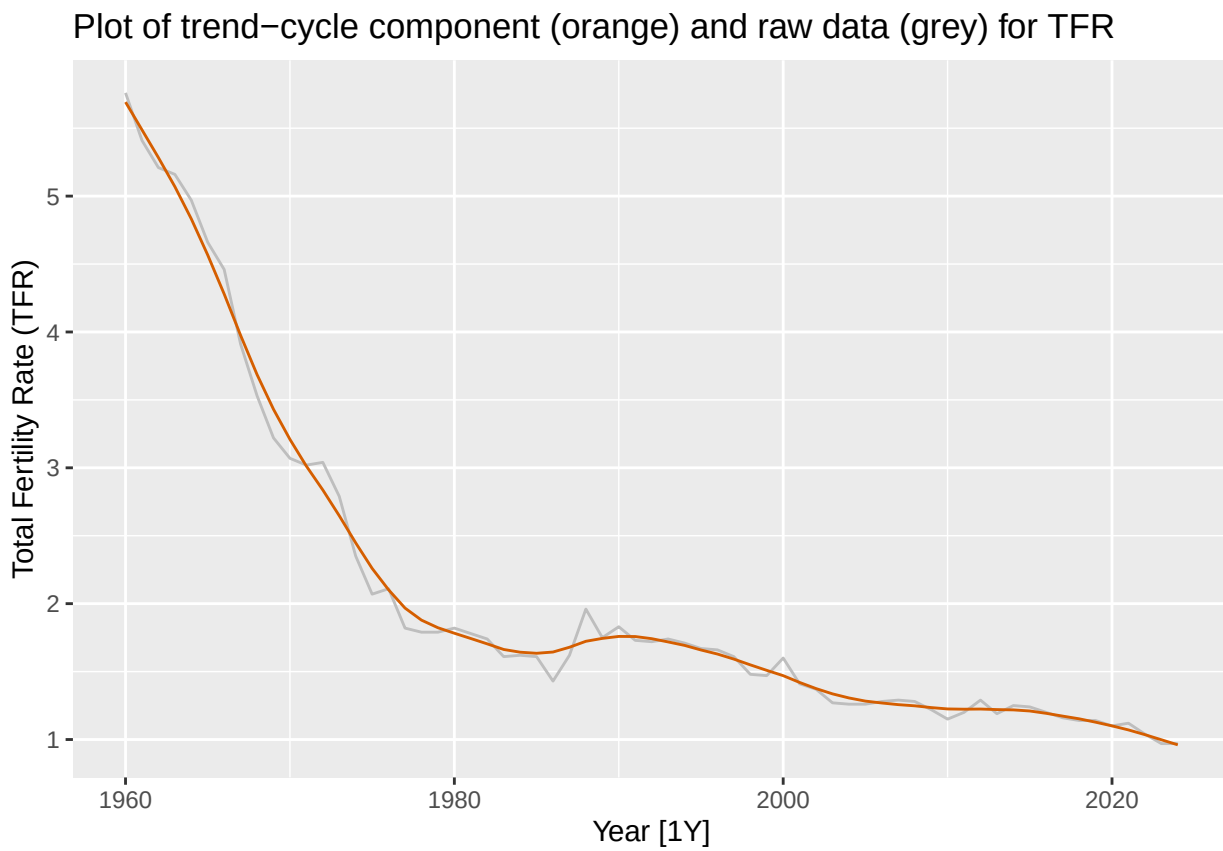
A Box-Cox transformation was considered, with the estimated lambda for TFR found to be below zero but approximately midway to -1, suggesting that a moderate transformation may be appropriate. The transformed series still shows a decreasing trend and there may also be cyclical behaviour; however the trend should be removed before confirming the presence of any cycles.

```r
lambdatlb <- BirthAndFertility |>
  features(`Total Live-Births`, features = guerrero) |> # Guerrero method
  pull(lambda_guerrero) # Box-Cox transformation parameter = -0.89992
BirthAndFertility |>
  autoplot(box_cox(`Total Live-Births`, lambdatlb)) +
  labs(title = latex2exp::TeX(paste0(
    "Transformed Total Live-Births (TLB) with $\\lambda$ = ", round(lambdatlb,2))))
```

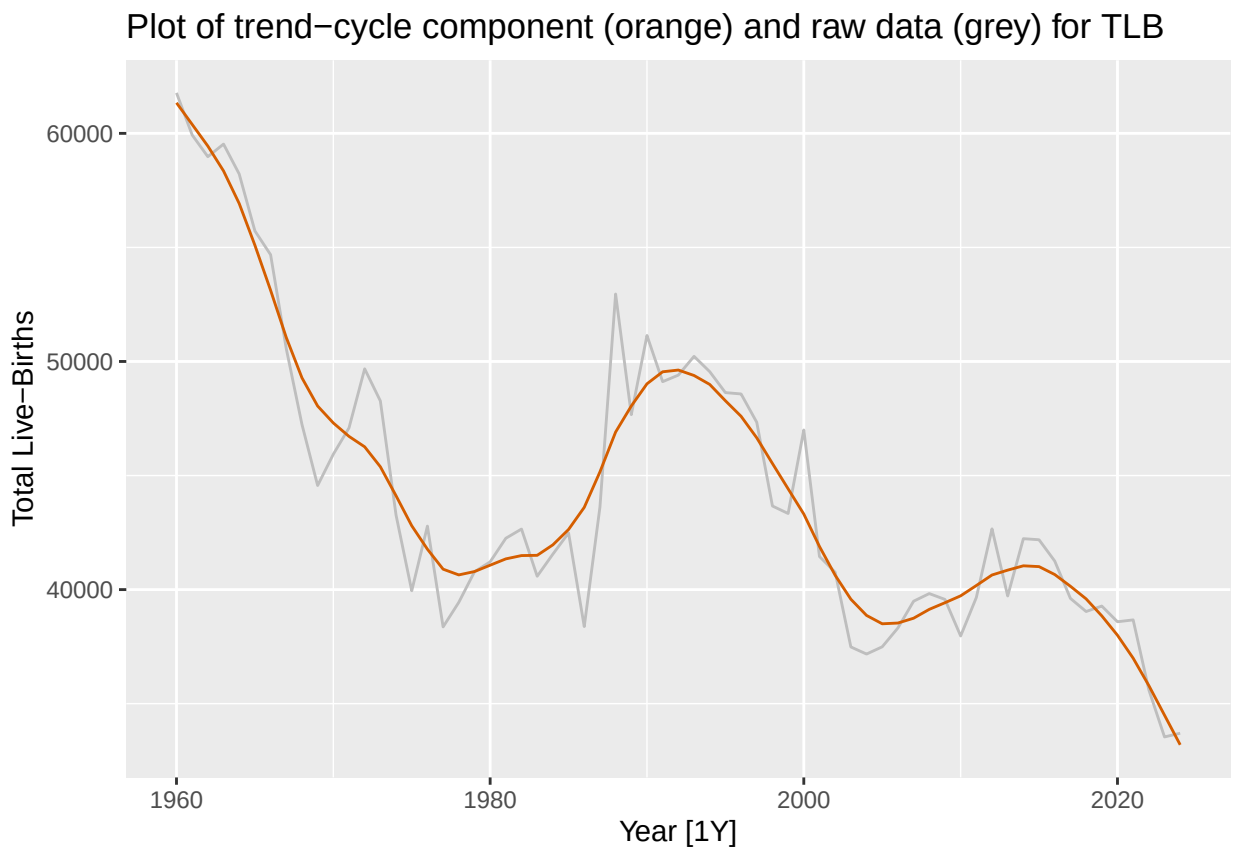


The estimated lambda for TLB was also found to be below zero but closer to -1, suggesting that a strong transformation may be required. The decreasing trend from 1960 to 2024 appears less prominent in this plot, and there might also be cyclical behaviour; however, the trend should be removed before confirming the presence of any cycles.

```r
dcmptfr <- BirthAndFertility |>
  model(stl = STL(`Total Fertility Rate (TFR)`))
components(dcmptfr) |>
  as_tsibble() |>
  autoplot(`Total Fertility Rate (TFR)`, colour = "grey") + # raw data
  geom_line(aes(y = trend), colour = "#D55E00") + # trend-cycle component
  labs(title = "Plot of trend-cycle component (orange) and raw data (grey) for TFR")
```

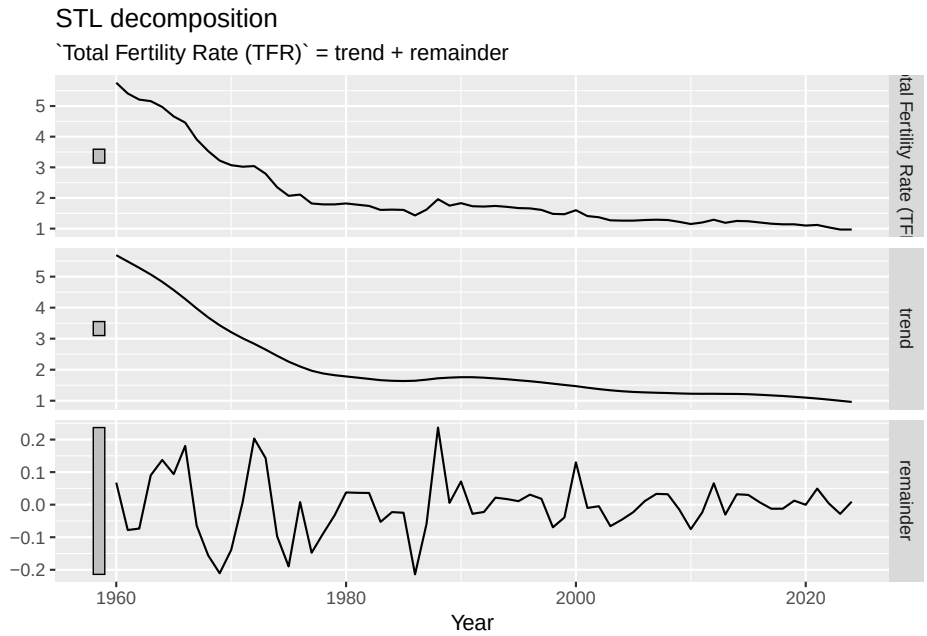


```r
dcmptlb <- BirthAndFertility |>
  model(stl = STL(`Total Live-Births`))
components(dcmptlb) |>
  as_tsibble() |>
  autoplot(`Total Live-Births`, colour = "grey") + # raw data
  geom_line(aes(y = trend), colour = "#D55E00") + # trend-cycle component
  labs(title = "Plot of trend-cycle component (orange) and raw data (grey) for TLB")
```

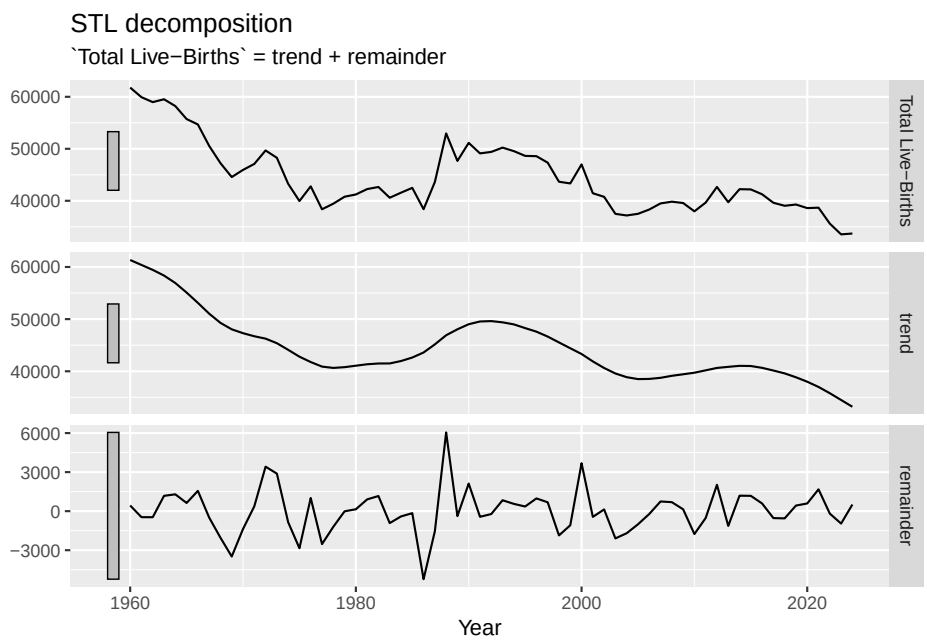


The STL decomposition suggests that both the TFR and TLB follow an additive structure with a clear trend component evident with the orange lines. The trend-component follows the TFR raw data smoothly, with little variability whereas the trend-component of TLB shows greater variability and larger fluctuations compared to its raw data.

```r
components(dcmptfr) |> autoplot()
```



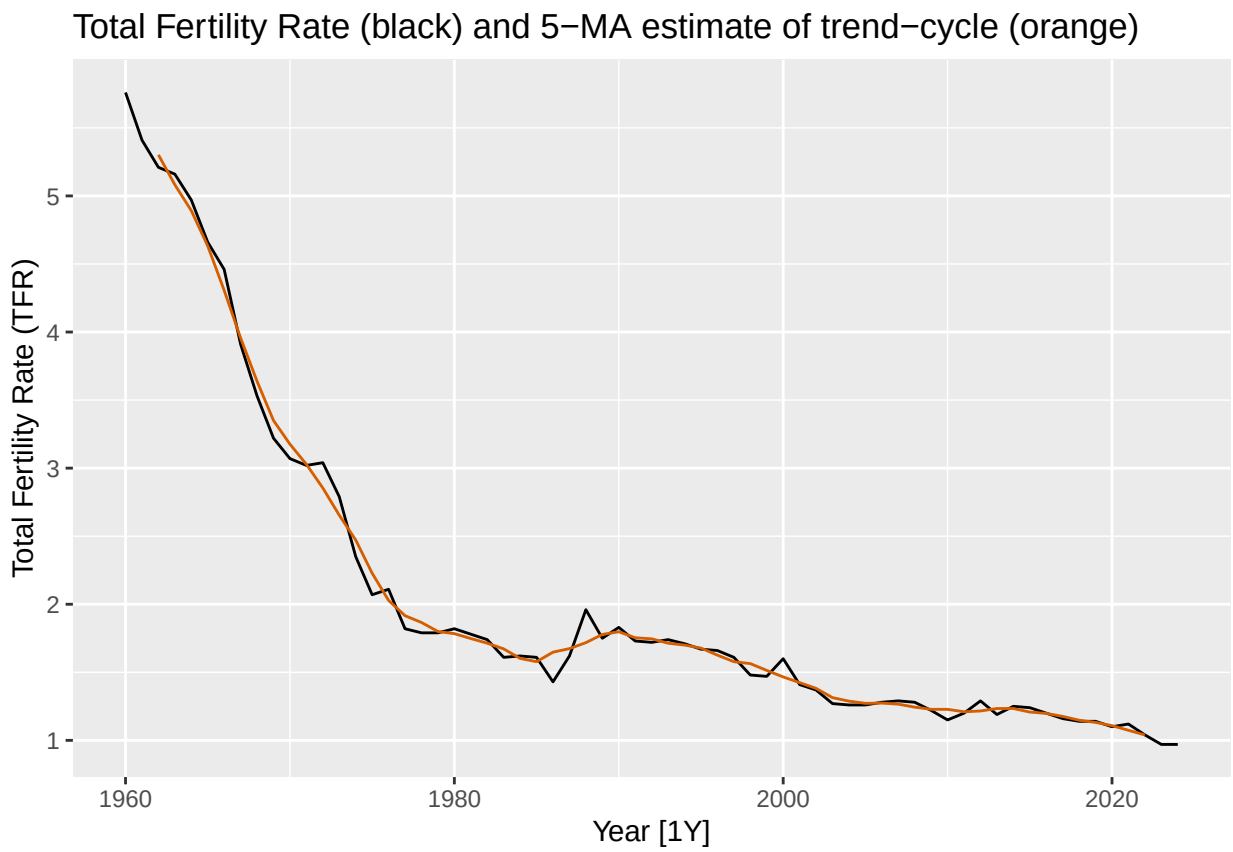

```r
components(dcmptlb) |> autoplot()
```



The STL decompositions for both TFR and TLB show an overall decreasing trend. However, there is a temporary increase beginning around 1980, reaching a broad peak around 1991 before decreasing again. The remainder components for both series display similar behaviour, with noticeable dip around 1986 followed by a spike around 1988.

```r
BirthAndFertility_MA_TFR <- BirthAndFertility |>
  mutate(`5-MA_TFR` = slider::slide_dbl(`Total Fertility Rate (TFR)`, mean,
                                        .before = 2, .after = 2, .complete = TRUE))
BirthAndFertility_MA_TFR |>
  autoplot(`Total Fertility Rate (TFR)`) +
  geom_line(aes(y = `5-MA_TFR`), colour = "#D55E00") +
  labs(title = "Total Fertility Rate (black) and 5-MA estimate of trend-cycle (orange)")
```

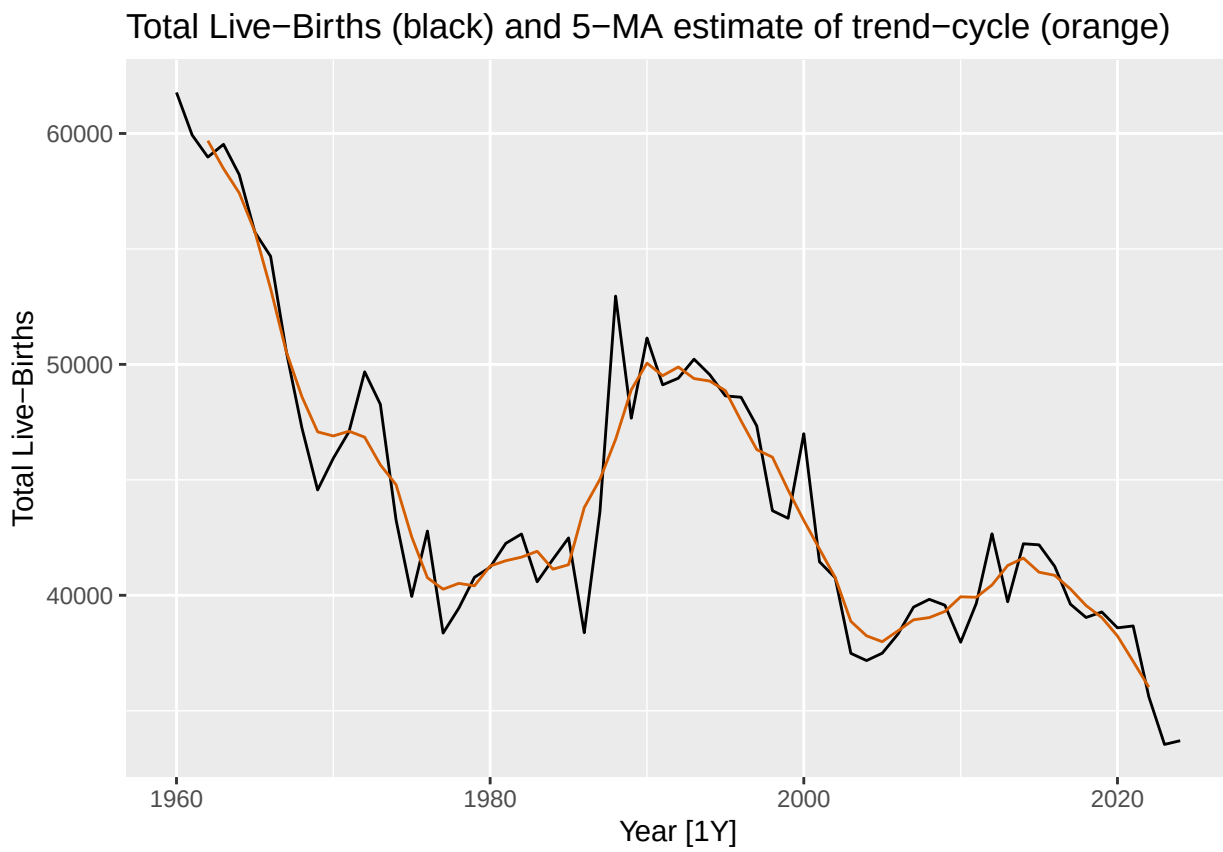
```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



The raw TFR series is already relatively smooth, so the 5-moving average closely follows the overall trend-cycle. However, it does fail to capture short-term fluctuations such as the dip around 1983 and spike around 1984.

```r
BirthAndFertility_MA_TLB <- BirthAndFertility |>
  mutate(`5-MA_TLB` = slider::slide_dbl(`Total Live-Births`, mean,
                                        .before = 2, .after = 2, .complete = TRUE))
BirthAndFertility_MA_TLB |>
  autoplot(`Total Live-Births`) +
  geom_line(aes(y = `5-MA_TLB`), colour = "#D55E00") +
  labs(title = "Total Live-Births (black) and 5-MA estimate of trend-cycle (orange)")
```

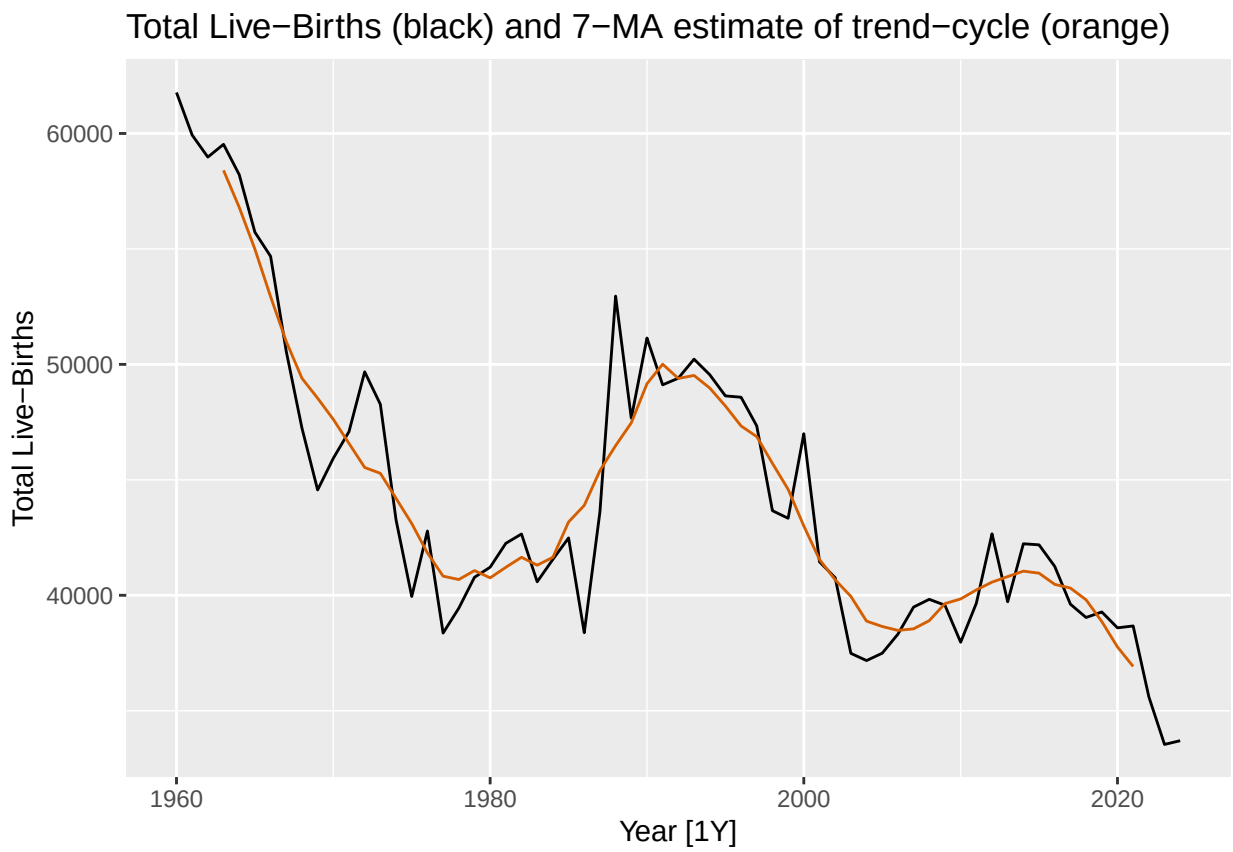
```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



The trend-cycle smooths out the short-term fluctuations in the raw TLB series. Increasing the window size of the current 5-moving average would result in a smoother curve.

```r
BirthAndFertility_MA_TLB7 <- BirthAndFertility |>
  mutate(`7-MA_TLB` = slider::slide_dbl(`Total Live-Births`, mean,
                                        .before = 3, .after = 3, .complete = TRUE))
BirthAndFertility_MA_TLB7 |>
  autoplot(`Total Live-Births`) +
  geom_line(aes(y = `7-MA_TLB`), colour = "#D55E00") +
  labs(title = "Total Live-Births (black) and 7-MA estimate of trend-cycle (orange)")
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



Evidently, the 7-moving average produces a smoother curve, at the cost of losing the observations at the ends of the series.

Statistical Appendix

The following statistical methods were applied throughout the analysis in the order presented below.

Autoregressive Model AR(p)

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

where:

y_t is the value at time t

ϕ_i are model parameters

ε_t is white noise

Dependence on using previous values (lags) to model

Autocorrelation Function (ACF)

$$\rho_k = \frac{\mathrm{Cov}(y_t, y_{t-k})}{\mathrm{Var}(y_t)}$$

where:

ρ_k is the autocorrelation at lag k

Measures the correlation between y_t and y_{t-k}

Residual

$$e_t = y_t - \hat{y}_t$$

y_t is the actual observed value at time t

$\hat{y}_t$ is the predicted value at time t

The difference between the actual observed value and predicted value at time t

STL Decomposition

$$y_t = T_t + R_t$$

where:

T_t is the trend component,

R_t is the remainder

Decomposes time series into components, usually with the seasonality component with the trend and remainder

Moving Average

$$MA_k = \frac{1}{k} \sum_{i=-(k-1)/2}^{(k-1)/2} y_{t+i}$$

k is the window size

In this analysis when k = 5, the average include the 2 points before the point at time t and 2 points after time t

Box-Cox Transformation

$$y^{(\lambda)} = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \lambda \neq 0 \\ \log(y), & \lambda = 0 \end{cases}$$

λ was estimated using the Guerrero method